

- 9 The standard enthalpy change of combustion of methane is  $-890 \text{ kJ mol}^{-1}$ .



Values of the relevant average bond energies are given in the table.

|     | average bond energy / $\text{kJ mol}^{-1}$ |
|-----|--------------------------------------------|
| C-H | 410                                        |
| C=O | 805                                        |
| O-H | 460                                        |
| O=O | 496                                        |

What is the value for the standard enthalpy change of vaporisation of water?



- A  $-72 \text{ kJ mol}^{-1}$     B  $-36 \text{ kJ mol}^{-1}$     C  $+36 \text{ kJ mol}^{-1}$     D  $+72 \text{ kJ mol}^{-1}$